

Rectangular Promotion: Cyclic Sieving, the Complete Cycle Ledger, and the Finite Koopman Spectrum

Route-A structural certificate C187

Abstract

For every rectangular standard Young-tableau set $X_{a,b} = \text{SYT}(b^a)$, we turn Rhoades's attributed cyclic-sieving theorem into a complete finite dynamical classification. Root-of-unity evaluations of the unshifted q -hook polynomial give every fixed iterate; Möbius inversion gives all least periods and cycles, hence the Artin–Mazur zeta, finite Koopman determinant, traces, and every spectral multiplicity. Evacuation reverses promotion. The theorem says $j^{ab} = 1$, not that the order always equals ab . The resulting finite unitary is natural but carries no intrinsic rational-prime semantics or target divisor.

Keywords: promotion; standard Young tableaux; cyclic sieving; q -hook formula; Koopman spectrum; evacuation.

中文摘要

本文对任意矩形标准杨表上的升运算给出完整有限动力学分类。以 Rhoades 的未平移 q 钩长循环筛分定理为归属明确的输入，单位根取值给出每个迭代的固定点数；Möbius 反演继而给出所有最小周期、循环、有限动力学 ζ 函数、Koopman 行列式、迹与谱重数。疏散算子将升运算共轭到其逆。统一结论是 $j^{ab} = 1$ ，而不是所有矩形的实际阶都等于 ab 。该有限酉算子虽属源系统自身，却没有内生的有理素数语义或目标除子匹配。

关键词：升运算；标准杨表；循环筛分； q 钩长公式；Koopman 谱；疏散。

1 Frozen convention and source theorem

Fix positive integers a, b , put $N = ab$, and let $X_{a,b} = \text{SYT}(b^a)$. We use the promotion convention j in Rhoades [1]: remove N , slide the hole northwest, increment the retained entries, and insert 1. Removing 1 and sliding southeast is the inverse clock and is not silently substituted.

For $[r]_q = (q^r - 1)/(q - 1)$ define the *unshifted* polynomial

$$F_{a,b}(q) = \frac{[N]_q!}{\prod_{c \in b^a} [h(c)]_q}.$$

Rhoades's rectangular standard-tableau CSP states, with $\omega_N = e^{2\pi i/N}$,

$$\# \text{Fix}(j^d) = F_{a,b}(\omega_N^d) \quad (d \in \mathbb{Z}). \quad (1)$$

The same source records the classical consequence $j^N = 1$, with order background attributed to Haiman [2]. It explicitly warns that the actual order need not be N . Thus the cyclic group of order N may act nonfaithfully.

The polynomial convention matters. The neighboring column-strict result in the source uses a shifted Schur specialization; importing that shift into (1) would change fixed counts. We use exactly the standard-tableau formula displayed above.

2 From fixed points to every cycle

Let P_ℓ be the number of tableaux of least promotion period ℓ , and let $C_\ell = P_\ell/\ell$. Since $j^N = 1$, only $\ell \mid N$ occur. A tableau is fixed by j^d precisely when its least period divides d , so divisor Möbius inversion gives

$$P_\ell = \sum_{d \mid \ell} \mu(\ell/d) F_{a,b}(\omega_N^d), \quad C_\ell = \frac{P_\ell}{\ell}. \quad (2)$$

This is an all-rectangle consequence of the source theorem, not an inference from a finite table. It also yields the actual promotion order as $\text{lcm}\{\ell : C_\ell > 0\}$, a divisor of N .

Shape	$ X_{a,b} $	nonzero cycle counts	actual order
$1 \times b$ or $a \times 1$	1	$C_1 = 1$	1
2×2	2	$C_2 = 1$	$2 < 4$
2×3	5	$C_2 = C_3 = 1$	6
3×3	42	$C_3 = 2, C_9 = 4$	9
3×4	462	$C_3 = 2, C_4 = 3, C_6 = 4, C_{12} = 35$	12

For the sharp boundary, $F_{2,2}(q) = 1 + q^2$. Its evaluations give no fixed tableau at one promotion step and two at the second, hence one two-cycle. A uniform claim of exact order N is therefore false even away from the one-dimensional singleton families.

3 Zeta, Koopman determinant, and reversal

The finite Artin–Mazur series and the cycle blocks give

$$\zeta_j(z) = \exp\left(\sum_{d \geq 1} \frac{\#\text{Fix}(j^d)}{d} z^d\right) = \prod_{\ell|N} (1 - z^\ell)^{-C_\ell}, \quad (3)$$

$$\det(I - zU_{a,b}) = \prod_{\ell|N} (1 - z^\ell)^{C_\ell} = \zeta_j(z)^{-1}, \quad (4)$$

where $(U_{a,b}f)(T) = f(jT)$ on counting $\ell^2(X_{a,b})$. Each cycle is a cyclic permutation block. Consequently

$$\text{Tr}(U_{a,b}^d) = \#\text{Fix}(j^d), \quad \text{mult}(\omega_N^k) = \sum_{\substack{\ell|N \\ N|k\ell}} C_\ell.$$

Summing these multiplicities over $k \bmod N$ gives $\sum_\ell \ell C_\ell = |X_{a,b}|$, so no spectral sector is missing.

Schützenberger evacuation e is an involution and satisfies $eje = j^{-1}$ on the rectangle [1]. Its permutation matrix E is a unitary reversor, $EUE = U^{-1}$. Since all matrices are real, E followed by coefficientwise conjugation is an antiunitary involution with the same reversal law. This is a natural source operator statement; it is not a post-hoc diagonal encoding.

4 Exact evidence and Route-A stop

The formula ledger covers all 36 rectangles with $1 \leq a, b \leq 6$: 441 iterate rows, 162 period rows, and 441 root-multiplicity rows. Its largest population is the 6×6 value 1,671,643,033,734,960, handled symbolically. An independent checker constructs 37,401 tableaux across 26 declared small rectangles and passes 230,034 assertions, including promotion/demotion, evacuation and all direct cycles. A separate SymPy path passes 3,065 checks; replay is byte exact; 107 repaired-hash and one stale-hash attacks are rejected.

These finite checks are regression evidence, not a new proof of the Kazhdan–Lusztig CSP. Source ownership remains with Rhoades and Haiman. The package contribution is the source-locked Route-A synthesis and executable boundary certificate.

Promotion has natural primitive cycles and a source-native finite unitary, but rectangle dimensions, hook lengths and cyclotomic orders do not intrinsically encode rational primes or prime powers. The finite root-of-unity determinant has no established target divisor, functional equation, continuation, counting law, or Weil compression. Therefore

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{WEAK}, \text{FAIL}, \text{FAIL}, \text{NATURAL QUANTIZATION}),$$

overall `ROUTE_A_REJECTED`, with Route B false and scope `NO_BAD_EULER_OR_ROOT_NUMBER`.

Revision-round focus. Round 1 adds the exact zeta, reciprocal Koopman determinant, trace identity, and complete root-of-unity multiplicity formula derived from the cycle blocks.

Limitations and nonclaims. We do not claim priority for cyclic sieving, the q-hook formula, the order bound, or evacuation. We do not claim exact order N uniformly or use finite enumeration as an infinite proof. No rational-prime semantics, target divisor, Hilbert–Polya operator, external review, or acceptance score is asserted.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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